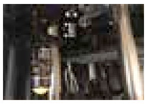
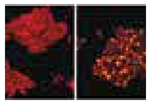


Shrink objects to nanoscale
MIT researchers have created a technique which allows the creation of 3D nanoscale objects of nearly any shape. **https://dgit.in/jan19-79**



Bacterial assassins?
Researchers have discovered a bacteria-killing virus that can eavesdrop on bacterial conversations and then attack diseases. **https://dgit.in/jan19-80**

Kepler-64b, an exoplanet with four suns was discovered by volunteers of the Planet Hunters project
Credit: JPL/Caltech

SPACE SCIENCE FOR AMATEURS

The universe is so vast that scientists just cannot study it all on their own, and there are many ways you can help.

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It has often happened that people have found things in the sky that had previously escaped the notice of professional astronomers. William Herschel, a music director, discovered an unknown planet, Uranus, in 1781. In 1951, Grote Reber, a radio operator built the first radio telescope and discovered two radio galaxies, Cygnus A and Cassiopeia A. Reber would go on to be considered as a pioneer in radio astronomy. In 1995, Thomas Bopp, a construction manager discovered the Hale-Bopp comet at the same time as Alan Hale, a professional in the field.

Now, because of the internet and household computers, scientists are actively seeking out the participation of the general public for accomplishing real scientific breakthroughs in space.

The astronomical instruments around the world and in space have together collected massive amounts of data. The data collected by the Cassini mission to Saturn, for example, is expected to be the basis of new scientific discoveries for at least another sixty years, even though the spacecraft has already disintegrated as a meteorite in the skies of Saturn. There are just not enough scientists around the world that can process all this data, which is why professional researchers have started reaching out to citizen scientists and amateur astronomers for help. The work does not involve lugging a telescope to a remote field and staring into space – but crunching numbers, sorting images, and looking for particular patterns in data.

DISTRIBUTED COMPUTING

One of the most well known and long running crowdsourced science projects is to scan signals from space for signs of intelligent life, known as the SETI@Home project. The project used the BOINC platform maintained by Berkeley, which is one of the leading distributed computing systems for solving science problems. Users can install the software on any operating system, and choose areas of interest to participate in, or pick particular projects. The Einstein@Home project lets you search for gravitational waves and look for signals from pulsars, or spinning neutron stars. The data is collected from the Arecibo radio